



WORK INTEGRATED LEARNING PROGRAMMES

NLP Applications

Assignment – 2

Problem Statement – 9

PART-B: Literature Survey

Group 55

2024AA05192	Abhishek Kumar Tiwari
2024AA05193	Krishanu Chakraborty
2024AA05197	Viswanadha Pavan Kumar
2024AA05832	B Vinod Kumar
2024AB05168	K Abhinav

Table of Contents

.....	1
TOPIC	3
INTRODUCTION.....	3
EVOLUTION OF SENTIMENT ANALYSIS	3
LIMITATIONS OF TEXT-BASED SENTIMENT ANALYSIS.....	3
How MULTIMODAL SENTIMENT ANALYSIS OVERCOME THE LIMITATION	3
CORE MODALITIES	4
TEXT MODALITY	4
VISUAL MODALITY	4
AUDIO MODALITY	4
FUSION TECHNIQUES IN MULTIMODAL SENTIMENT ANALYSIS.....	4
EARLY FUSION (FEATURE-LEVEL).....	4
LATE FUSION (DECISION-LEVEL)	4
HYBRID & TRANSFORMER-BASED FUSION (STATE-OF-THE-ART)	4
APPLICATIONS OF MULTIMODAL SENTIMENT ANALYSIS.....	5
POPULAR BENCHMARK DATASETS	5
CHALLENGES & FUTURE DIRECTIONS	5
CONCLUSION	5

Topic

Sentiment Analysis on Multimodal Data

Introduction

Sentiment Analysis also known as opinion mining which is a Natural Language Processing task that aims to determine the emotional tone behind textual content. Traditionally sentiment analysis has focused on text-only data such as reviews, tweets, and blogs.

However, modern digital communication platforms like YouTube, Instagram increasingly involve multimodal content, where text is combined with images, videos, and audio. In such contexts, sentiment is expressed not only through words but also through facial expressions, voice tone, gestures and body language, visual context in images.

Multimodal Sentiment Analysis is a subfield of NLP and Computer Vision that aims to integrate these heterogeneous data streams to achieve a more robust and human-like understanding of emotions. which integrates multiple data modalities to improve sentiment prediction accuracy.

Evolution of Sentiment Analysis

Early sentiment analysis systems relied on lexicon-based approaches, machine learning models, deep learning models, which includes NLTK, SPACY. While effective for structured text, these models struggle when sentiment depends on contextual cues beyond text such as sarcasm or emotional tone.

Limitations of text-based sentiment analysis

Text alone cannot fully capture the richness of human communication, Elements such as tone of voice, facial expressions, and visual cues play a crucial role in conveying emotion, intent, and nuance. Without these, messages can easily be misunderstood, especially when it comes to sarcasm, which often relies on expressions, gestures, or emphasis rather than words themselves. As a result, purely text-based communication may lack clarity and emotional depth compared to face-to-face interactions

How Multimodal Sentiment Analysis Overcome the Limitation

Multimodal Sentiment Analysis is an advanced approach to sentiment detection that analyzes emotions by combining multiple data modalities instead of relying on a single source. It integrates textual information such as transcripts and captions to understand linguistic sentiment, audio cues like pitch, tone, and MFCCs to capture emotional variations in speech, and visual signals including facial expressions and gestures to interpret non-verbal emotions. Additionally, images such as memes and emojis contribute contextual and emotional meaning, while video data helps analyze body language and overall behavioral patterns. By fusing these modalities, multimodal sentiment analysis provides a more accurate and holistic understanding of human emotions and intent.

Core Modalities

Multimodal Sentiment Analysis combines information from multiple data sources:

Text modality

Which is used for analyzing words and linguistic features, Text remains the core modality and is processed using tokenization, embeddings (Word2Vec, BERT), transformers. Modern systems use models such as BERT, Transformers. Text provides explicit sentiment signals but may miss implicit emotional cues.

Visual modality

Which is used for analyzing Facial expressions, scene context. Visual sentiment is extracted from Facial expressions, Object presence, Scene understanding, Body posture. Deep learning models like Convolutional Neural Networks (CNNs) are used to extract visual features. Visual cues help detect sarcasm or contrast example like smiling while saying something negative.

Audio modality

Which is used for analyzing Voice pitch, tone, intensity. Audio sentiment analysis extracts pitch variation, speech rate, energy levels, tone. Then Models convert speech signals into Spectrograms, MFCC features. Recurrent Neural Networks (RNNs) and Transformers process temporal speech patterns.

Fusion Techniques in Multimodal Sentiment Analysis

The fusion of these modalities is the most critical technical challenge. Literature classifies fusion into three main approaches:

Early Fusion (Feature-level)

Modality features are concatenated into a single high-dimensional vector before being fed into a classifier.

Benefit: Captures low-level correlations early.

Drawback: Struggles with "modality redundancy" and synchronization.

Late Fusion (Decision-level)

Each modality is processed by a separate model example as CNN for images, LSTM for text and their individual predictions are combined via averaging or voting at the final stage.

Benefit: Robust to missing modalities.

Drawback: Ignores cross-modal interactions example of sarcasm where text is positive, but tone is angry.

Hybrid & Transformer-based Fusion (State-of-the-Art)

Recent research heavily utilizes Cross-Modal Attention Mechanisms. Models like Multimodal Transformer or Mamba-driven networks allow one modality to "attend" to another, aligning features dynamically.

Recent state of the art research integrates Multimodal Transformers, Attention Mechanisms, Graph Neural Networks, Influential research trends include, Cross-modal transformers, Contrastive learning, Multimodal large language models.

Applications of Multimodal Sentiment Analysis

Multimodal sentiment analysis is widely applied in social media monitoring, customer experience analysis, video content recommendation, political opinion mining, mental health monitoring, human-computer interaction.

Popular Benchmark Datasets

A literature survey is incomplete without mentioning the datasets that drive the field:

CMU-MOSI/CMU-MOSEI: The gold standard for video-based sentiment and emotion recognition.

MELD: Multi-party Emotion Dataset (based on the TV show Friends), used for conversational context.

MuSe-CaR: Specifically for user-generated video reviews (e.g., car unboxings).

Challenges & Future Directions

Modality Alignment: Synchronizing a text token with a specific video frame or audio millisecond.

Sarcasm & Context: The primary advantage of MSA; research shows that visual cues are often the only way to detect "deception" in positive text.

Explainability: Understanding why a model gave a negative score example sharp tone and the specific word.

Resource Efficiency: Moving away from heavy architectures to lightweight models like Mobile Net backbones for real-time mobile applications.

Data Scarcity: Multimodal datasets are expensive to collect and annotate.

Conclusion

Multimodal Sentiment Analysis represents a significant advancement over traditional text-based sentiment analysis by integrating information from multiple modalities such as text, audio, and visual data. By leveraging technologies from natural language processing, speech signal processing, and computer vision, and combining them through advanced fusion techniques like attention mechanisms and multimodal transformers, multimodal sentiment analysis enables a richer and more accurate understanding of human emotions and opinions.

This approach is particularly effective in handling complex emotional expressions, sarcasm, and contextual nuances that cannot be reliably captured using a single modality. Although multimodal systems are computationally more complex and require carefully aligned and high-quality data, they offer superior performance and robustness in real-world applications such as human-computer interaction, social media analysis, customer feedback analysis, and affective computing.

In conclusion, multimodal sentiment analysis provides a more human-like and context-aware sentiment understanding, making it a crucial research and application area as intelligent systems increasingly interact with diverse and expressive forms of human communication.